Name: ____________________________ Date: ________________ Period: _________

Spiral Review — Key

1. Let $f(t) = 3t^2 - 12t$.

(a) Is f increasing or decreasing on the interval $[0, 2]$? How do you know? (Hint: compare $f(0)$ and $f(2)$.)

$f(0) = 0$ and $f(2) = 3(4) - 24 = -12$. Since $f(2) < f(0)$, the function is **decreasing** on $[0, 2]$.

(b) Is f increasing or decreasing on the interval $[2, 5]$?

$f(2) = -12$ and $f(5) = 75 - 60 = 15$. Since $f(5) > f(2)$, the function is **increasing** on $[2, 5]$.

2. Compute the average rate of change of $g(t) = t^2 - 4$ on the interval $[1, 4]$. Show your work.

$g(1) = 1 - 4 = -3$ and $g(4) = 16 - 4 = 12$.

$$\text{ARC} = \frac{g(4) - g(1)}{4 - 1} = \frac{12 - (-3)}{3} = \frac{15}{3} = 5$$

3. Two particles both visit the point $(3, 5)$ in the plane. Particle A arrives when $t = 1$; Particle B arrives when $t = 2$. Explain why this is possible for parametric functions but is *not* possible for a function $y = f(x)$.

In a parametric function, two different values of the parameter t can produce the same ordered pair (x, y) at different times. The parameter t tracks *when* the particle is at a location, not just where. In a function $y = f(x)$, each input $x = 3$ produces exactly one output y, so only one particle can be at $x = 3$ with a given y-value — the timing is not encoded.

Teacher Note

Q1 previews the direction-of-motion analysis: students will use a similar comparison of values (rather than derivatives) to determine whether $x(t)$ or $y(t)$ is increasing or decreasing. Q2 directly activates the average rate of change computation used in EK 4.3.A.4. Q3 surfaces the key conceptual advantage of parametric functions over $y = f(x)$: two different t-values can map to the same (x, y) point, which EK 4.3.A.2 formalizes.